

Technology Stack

Date	11 July 2026
Team ID	XXXXXX
Project Name	EduGenie – Google Gemini Powered Learning Assistant
Maximum Marks	2 Marks

Define Your Technology Stack

EduGenie – Google Gemini Powered Learning Assistant is developed using a modern and scalable technology stack consisting of frontend technologies, backend frameworks, AI models, databases, deployment platforms, and third-party APIs. These technologies were selected to provide high performance, maintainability, scalability, security, and an interactive learning experience for students and educators.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, JavaScript	Provides a responsive and user-friendly interface for students and teachers to interact with EduGenie through forms, buttons, and dynamic content.
2	Backend / Server-Side	Python, FastAPI, Uvicorn	Handles application logic, API development, request processing, AI integration, and response generation with high performance.
3	AI / Large Language Models	Google Gemini 1.5 Pro, LaMini-Flan-T5	Performs concept explanation, question answering, quiz generation, text summarization, and personalized learning recommendations.
4	Database / Data Storage	JSON Files, Local Storage	Stores user preferences, generated quizzes, summaries, learning history, and application configuration in a lightweight format.
5	Cloud / Deployment	Google AI Studio, Uvicorn (Local Deployment)	Provides access to the Gemini API and enables local deployment of the EduGenie application for development and testing.
6	Version Control & Collaboration	Git & GitHub	Supports version control, collaboration among team members, project tracking, and source code management.
7	Third-Party APIs / Libraries	Google Generative AI SDK, Transformers, Hugging Face, Pydantic, Jinja2	Enables AI model integration, natural language processing, request validation, template rendering, and efficient backend development.